Question	Answer	Marks	Partial Marks
16	$\frac{65}{198}$ cao	**3**	**B2** for a fraction equivalent to $\frac{65}{198}$ e.g. $\frac{325}{990}$ or **M1** for $328.\dot{2}\dot{8} - 3.\dot{2}\dot{8}$ oe or $\frac{3}{10} + \frac{28}{990}$ oe or $990x = 325$ or $\frac{k}{990}$
17	21	**3**	**M2** for $7 \times \sqrt{\frac{540}{60}}$ oe or $7 \div \sqrt{\frac{60}{540}}$ oe or **M1** for $\sqrt{\frac{540}{60}}$ oe or $\sqrt{\frac{60}{540}}$ oe or $\left(\frac{7}{H}\right)^2 = \frac{60}{540}$ oe
18	$[t =]\frac{m}{2p+m}$ final answer	**4**	**M1** for correctly clearing *their* fraction **M1** for correct expansion **M1** for correctly collecting *their* terms in t on one side and other terms on the other side, not dividing by $1 - t$ **M1** for correct factorisation and division of *their* two-term expression in t To a maximum of 3 marks for an incorrect answer
19	$\frac{x}{7+x}$ final answer	**3**	**B1** for $x(7 - x)$ **B1** for $(7 - x)(7 + x)$
20	$\frac{32}{3}\pi$ or $10\frac{2}{3}\pi$ cao	**3**	**M2** for $\frac{360-40}{360} \times \pi \times 6 \times 2$ or $\frac{360-40}{360} \times \frac{22}{7} \times 6 \times 2$ oe or **M1** for $\frac{40}{360} \times \pi \times 6 \times 2$ oe or **B1** for major sector angle = 320 soi or 12π
21(a)	$3x^2 - 6x$ or $3x(x - 2)$	**2**	**B1** for $3x^2 - kx$ or $kx^2 - 6x$ or correct answer spoilt or **SC1** for $3x^2 - 6x + 1$

Question	Answer	Marks	Partial Marks
21(b)	(0, 1) (2, –3)	**4**	**B3** for one correct coordinate or for two correct values of x following $3x^2 - 6x$ oe seen in (b) or **M2** for [3]$x(x - 2)$ [= 0] or $\frac{--6 \pm \sqrt{(-6)^2 - 4 \times 3 \times 0}}{2 \times 3}$ [= 0] oe OR **M1** for *their* $\frac{dy}{dx} = 0$ or stating $\frac{dy}{dx} = 0$ **M1** for correct method to solve *their* quadratic
22(a)	11	**1**	
22(b)	$\frac{x-5}{2}$	**2**	**M1** for correct first step $x = 2y + 5$ or $y - 5 = 2x$ or $\frac{y-5}{2} = x$ or $\frac{y}{2} = x + \frac{5}{2}$
22(c)	14	**3**	**M2** for $x - 4 = (25 - 5) \div 2$ oe or better or $2x - 8 = 25 - 5$ oe or better or **M1** for $2(x - 4) + 5 = 25$
22(d)	25	**2**	**M1** for h(2) or 5^2
23(a)	0	**1**	